



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

CHEMISTRY

PAPER 1 Multiple Choice
SPECIMEN PAPER

4024/1

1 hour

Additional materials:

- Multiple Choice answer sheet
- Soft clean eraser
- Soft pencil (type B or HB is recommended)
- Electronic calculator

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

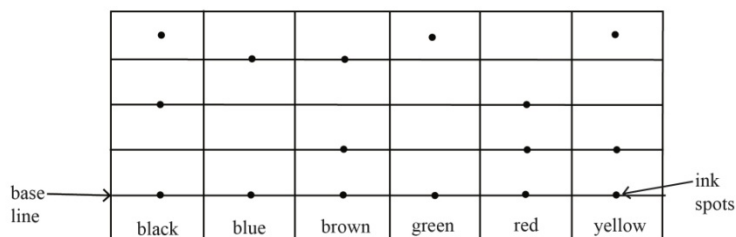
Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer. Any rough working should be done in this booklet. A copy of the Periodic Table is printed on page 15.

This specimen paper consists of 15 printed pages.
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1. The smell of perfume in a room spreads quicker on a hot day than on a cold day because the particles
 - A become lighter.
 - B become heavier.
 - C have more kinetic energy.
 - D have less kinetic energy.

2. Which instrument accurately measures 17.50 cm^3 of water?
 - A measuring cylinder
 - B burette
 - C pipette
 - D beaker

3. The diagram shows a chromatogram of different inks.



Which statement about the chromatogram is **not** true?

- A brown ink contains dyes in red and blue ink only
- B green and yellow ink contain a common dye
- C black ink contains a common dye with red ink
- D the dyes in red ink are not found in blue ink

4. The table shows pH range values of some natural substances.

substance	pH range
lemon juice	4-5
bee sting	5-7
wood ash	8-9
ice blocks	7
mango juice	4.5-6

Which substance is most suitable for relieving pain from a bee sting?

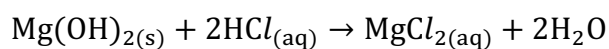
- A lemon juice
 - B wood ash
 - C ice blocks
 - D mango juice
5. Which method is used to separate a mixture of hydrocarbons of different boiling points?
- A evaporation.
 - B simple distillation.
 - C fractional distillation
 - D decanting.

6. The table shows information about ethanol and sulphur.

	melting point / °C	boiling point / °C
ethanol	- 15	78
sulphur	119	445

Which statement is true about the two substances?

- A ethanol is a solid at room temperature
 - B sulphur is a solid at room temperature
 - C ethanol is a gas at room temperature
 - D sulphur is a gas at room temperature
7. The active ingredient, $\text{Mg}(\text{OH})_2$, in indigestion tablets reacts with excess acids in the stomach according to the equation:



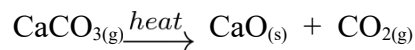
What is the effect, on the rate of reaction, of using the tablets in powder form?

- A increases
 - B decreases
 - C no effect
 - D decreases then increases
8. Which statement is true about the electronic structure of atoms?
- A outer shells of different atoms have the same number of electrons
 - B inner shells of different atoms have the same number of electrons
 - C electrons enter into different shells across a period.
 - D electrons enter into the same shell across a period.

9. Which substance conducts electricity when in solid state?
- A diamond
 - B graphite
 - C sodium chloride
 - D magnesium oxide
10. Which chemical equation represents the final stage in the production of nitric acid ?
- A $NO_2 + H_2O \longrightarrow H_2NO_3$
 - B $NH_3 + O_2 \longrightarrow NO + H_2O$
 - C $4NO_2 + 2H_2O + O_2 \longrightarrow 4HNO_3$
 - D $NO + O_2 + H_2O \longrightarrow HNO_3 + OH^-$
11. Alcohols are formed when alkenes react with
- A water.
 - B steam.
 - C hydrogen.
 - D sodium hydroxide.
12. What is the role of aluminium sulphate in the water purification process?
- A chlorination
 - B coagulation
 - C softening
 - D flocculation

13. Catalytic converters reduce the emission of the following **except**
- A carbon monoxide.
 - B carbon dioxide.
 - C nitrogen oxides.
 - D unburnt hydrocarbons.
14. Nitric acid was added to solution X, followed by aqueous silver nitrate. A white precipitate was obtained.
- Which anion could be present in solution X?
- A NO_3^-
 - B SO_4^{2-}
 - C Cl^-
 - D OH^-
15. Atoms represented by $^{35}_{17}\text{Cl}$ and $^{37}_{17}\text{Cl}$ are examples of
- A isomers.
 - B isotopes.
 - C alloys.
 - D allotropes.

16. When calcium carbonate is heated, it decomposes according to the equation:



What is the volume of carbon dioxide produced when 5 g of calcium carbonate were heated?

[molar gas volume at r.t.p = 24 dm³]

- A 2.73 dm³
 B 1.76 dm³
 C 1.20 dm³
 D 0.81 dm³
17. A volume of 50 cm³ of 0.1 moldm⁻³ sodium hydroxide were completely neutralised by 25 cm³ of 0.1 moldm⁻³ sulphuric acid. The heat change for the reaction was 1575 J.

The enthalpy of neutralisation is

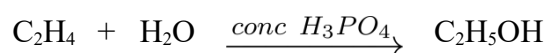
- A 630.000 kJmol⁻¹.
 B 315.000 kJmol⁻¹.
 C 105.000 kJmol⁻¹.
 D 1.575 kJmol⁻¹.

18. Which statement(s) 1, 2 or 3 refer to elements in the same group?
- (i) same number of valence electrons
 - (ii) similar chemical properties
 - (iii) same colour and reactivity
- A 3 only
- B 2 and 3
- C 1, 2 and 3
- D 1 and 2
19. The effect of global warming on the environment is that it causes an increase in
- A ozone depletion.
- B eutrophication.
- C atmospheric temperature.
- D oxygen in the atmosphere.
20. The molecular formula of an organic alcohol that contains 60% carbon by mass is
- A CH_3OH .
- B $\text{C}_2\text{H}_5\text{OH}$.
- C $\text{C}_3\text{H}_7\text{OH}$.
- D $\text{C}_4\text{H}_9\text{OH}$.

21. Which pair of reactants are used to form potassium nitrate?

	acid	base
A	nitric acid	sodium hydroxide
B	hydrochloric acid	potassium hydroxide
C	nitric acid	potassium hydroxide
D	hydrochloric acid	sodium hydroxide

22. The equation shows the formation of ethanol from ethene using steam.



What temperature is used in the reaction?

- A 300°C
 B 450 °C
 C 700 °C
 D 900 °C
23. The table shows the changes that occur when halogens react with different potassium halides.

	<i>KCl</i>	<i>KBr</i>	<i>KI</i>
<i>Cl</i> ₂	no change	turns orange	turns brown
<i>Br</i> ₂	no change	no change	turns brown
<i>I</i> ₂	no change	no change	no change

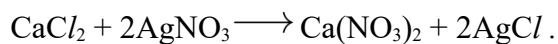
Which statement explains the changes in colour?

- A chlorine and bromine displace iodine
 B bromine displaces iodine and chlorine
 C iodine and chlorine displace bromine
 D iodine displaces chlorine and bromine

24. During the liquefaction and fractional distillation of air, air is first cooled to $-78\text{ }^{\circ}\text{C}$ because
- A carbon dioxide solidifies.
 - B rare gases solidify.
 - C oxygen liquefies
 - D nitrogen liquefies.
25. A floatation cell is used to concentrate the ore in the extraction of
- A nickel.
 - B copper.
 - C diamonds.
 - D iron.
26. An object is electroplated with copper.
- Which statement about the process is correct?
- A oxygen is evolved at the anode
 - B copper ions gain electrons at the object being electroplated
 - C copper is used as the cathode
 - D the concentration of copper ions in the electrolyte decreases
27. Which pair of metals gives the highest voltage reading when used in a simple cell?
- A Mg and Fe
 - B Mg and Cu
 - C Zn and Pb
 - D Fe and Cu

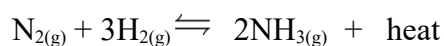
28. Which reagents are used in the production of ammonium nitrate fertiliser?
- A HNO_3 , NH_3
 - B NO , NH_3
 - C HNO_2 , N_2H_4
 - D NO_2 , NH_4OH
29. Which one is a correct physical property of solid ionic compounds?
- A low boiling point and melting point
 - B high boiling point and melting point
 - C good electrical conductors
 - D highly soluble in organic solvents
30. Iron can be extracted from its ore by direct reduction using carbon monoxide whereas aluminium can only be extracted by electrolysis because
- A iron is more reactive than aluminium and more difficult to extract.
 - B aluminium is more reactive than iron and more difficult to extract.
 - C aluminium is common and its extraction must be simple.
 - D iron is common and its extraction must be simple.
31. Which oxide reacts with both hydrochloric acid and sodium hydroxide?
- A CaO
 - B NaO
 - C Al_2O_3
 - D NO_2

32. A mass of 8 grams of calcium chloride reacted with silver nitrate to produce a white precipitate of silver chloride. The chemical equation for the reaction is



The mass of silver chloride produced is

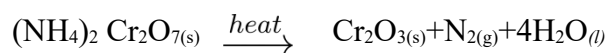
- A 6.80 g
B 10.40 g
C 13.60 g
D 20.70 g
33. Which chemical equation shows how barium reacts with water?
- A $\text{Ba} + 2\text{H}_2\text{O} \longrightarrow \text{Ba}(\text{OH})_2 + \text{H}_2$
B $\text{Ba} + \text{H}_2\text{O} \longrightarrow 2\text{BaO} + \text{H}_2$
C $\text{Ba} + \text{H}_2\text{O} \longrightarrow \text{BaO} + \text{H}_2$
D $\text{Ba} + \text{H}_2\text{O} \longrightarrow \text{Ba}(\text{OH})_2 + \text{H}_2$
34. Ammonia can be produced industrially from nitrogen and hydrogen. The equation for the reaction is



Which change would increase the yield of ammonia during the reaction?

- A adding a catalyst.
B decreasing the pressure.
C increasing the temperature.
D increasing the pressure.

35. Ammonium dichromate, $(\text{NH}_4)_2 \text{Cr}_2 \text{O}_7(\text{s})$ decomposes on heating to form chromium (III) oxide, Cr_2O_3 , nitrogen gas and water according to the equation:



When a sample containing 50.40 g of $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ was heated, 4.32 g of water were produced.

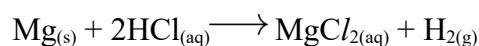
What was the percentage purity of the ammonium dichromate?

- A 30.00%
- B 9.00%
- C 0.30%
- D 0.09%
36. Which reagent can be used to distinguish between propene and propane?
- A ethanol
- B aqueous bromine
- C aqueous ammonia
- D dilute sulphuric acid
37. Ethane reacts with chlorine to form chloro-ethane in the presence of
- A an aluminium oxide.
- B concentrated sulphuric acid.
- C ultraviolet light.
- D bromine solution.

38. Which compound is formed when an organic acid reacts with an alcohol?

- A $\text{CH}_3\text{CH}_2\text{OH}$
- B CH_3COOH
- C $\text{CH}_3\text{COOCH}_3$
- D CH_3COCH_3

39. Magnesium metal reacts with hydrochloric acid as follows:



What mass of magnesium will produce 36 dm^3 of hydrogen at room temperature and pressure (r.t.p)?

- A 24 g
 - B 36 g
 - C 48 g
 - D 72 g
40. Which equation shows a balanced chemical equation for the reaction of barium and sulphuric acid?

- A $\text{Ba}_{(s)} + \text{H}_2\text{SO}_{4(aq)} \longrightarrow \text{Ba}(\text{SO}_4)_{2(s)} + 2\text{H}_{2(g)}$
- B $\text{Ba}_{(s)} + \text{H}_2\text{SO}_{4(aq)} \longrightarrow \text{BaSO}_{4(s)} + \text{H}_{2(g)}$
- C $\text{Ba}_{2(s)} + \text{H}_2\text{SO}_{4(aq)} \longrightarrow \text{Ba}_2\text{SO}_{4(s)} + \text{H}_{2(g)}$
- D $\text{Ba}_{(s)} + 2 \text{HSO}_{4(aq)} \longrightarrow \text{BaSO}_{4(s)} + \text{H}_{2(g)}$

Group

*58-71 Lanthanoid series
†90-103 Actinoid series

[illegible]